

ATUL SHARMA

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[LinkedIn](#) | [Google Scholar](#) | [Website](#)

RESEARCH STATEMENT

My research focuses on collaborative machine learning (CML), where multiple nodes jointly train models without sharing raw data. My early work studied byzantine-robust aggregation in federated learning, where collaboration is coordinated through a central server. I later explored improving model mixing and convergence under communication constraints in distributed settings. My current work investigates fully decentralized learning protocols that allow nodes to autonomously discover and collaborate with peers with similar data distributions, improving personalization in non-IID environments.

EDUCATION

Purdue University, College of Engineering West Lafayette, IN
Ph.D. in Electrical and Computer Engineering, GPA – 3.8/4.0 *Aug 2019 - Aug 2026*

Indian Institute of Technology Kharagpur West Bengal, India
B.Tech. Major in E&ECE (Honors), GPA – 9.4/10; Minor in CSE, GPA – 9.3/10 *Jul 2015 - May 2019*

INTERNSHIP EXPERIENCE

Graduate Internship

Google LLC Sunnyvale, CA
Software Engineering Intern *Jun 2023 - Aug 2023*

- Trained ML models on telemetries to estimate **SSD flash wear-out**, extending hardware lifespan in Google data centers.
- Engineered an end-to-end data pipeline achieving 80% precision and 90% recall in predicting critical bad block growth.

NVIDIA Graphics Pvt Ltd Santa Clara, CA
Applied Power Architecture Intern *Jun 2022 - Aug 2022*

- Implemented **Implicit Q-Learning (IQL)** to automate CPU/GPU operating point selection under power limits.
- Achieved **16x inference speedup** with no accuracy loss by replacing existing ML models with an optimized RL agent.

NVIDIA Graphics Pvt Ltd Santa Clara, CA
Applied Power Architecture Intern *May 2021 - Aug 2021*

- Extended **Dynamic Boost 2.0** to non-graphics workloads to optimize power allocation between the CPU and GPU.
- Developed a **sequence alignment algorithm** to identify identical workloads across operating points, creating a high-fidelity dataset for an ML model trained to predict performance and optimize power juggling between the CPU and GPU.

Undergraduate Internship

NVIDIA Graphics Pvt Ltd Bengaluru, India
ASIC Engineer *May 2018 - Jul 2018*

- Developed a generic sequence monitor and a custom testbench to verify **power gating features in GPUs**.
- Integrated and deployed the sequence monitor into GPU fullchip testbench and verified its functionality.

AWARDS

Silver Reviewer Designation, *International Conference on Machine Learning (ICML) 2026*

- Recognized as a top-tier reviewer by Area Chairs based on evaluation quality and rigor.

PATENTS & PUBLICATIONS

-
- Saurabh Bagchi, Somali Chatterji, Atul Sharma. "System and Methods for Collaborative Federated Learning." U.S. Patent Application No. 19/251,710 (Published: US20260004192A1), Jan 2026. [[Patent](#)]

- Zhao, Joshua C., Ahmed Roushdy Elkordy, *Atul Sharma*, Yahya H. Ezzeldin, Salman Avestimehr, and Saurabh Bagchi. "LOKI: Large-scale Data Reconstruction Attack against Federated Learning through Model Manipulation." In Proceedings of the 2024 IEEE Symposium on Security and Privacy. [\[Article\]](#)
- *Atul Sharma*, Wei Chen, Joshua Zhao, Qiang Qiu, Saurabh Bagchi, and Somali Chaterji. "How to Learn Collaboratively - Federated Learning to Peer-to-Peer Learning and What's at Stake." 2023 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks - Supplemental Volume (DSN-S). [\[Article\]](#)
- *Atul Sharma*, Wei Chen, Joshua Zhao, Qiang Qiu, Saurabh Bagchi, and Somali Chaterji. "FLAIR: Defense against Model Poisoning Attack in Federated Learning." 2023 ACM Asia Conference on Computer and Communications Security (ASIA CCS '23). [\[Article\]](#)
- Zhao, Joshua C., Ahmed Roushdy Elkordy, *Atul Sharma*, Yahya H. Ezzeldin, Salman Avestimehr, and Saurabh Bagchi. "The Resource Problem of Using Linear Layer Leakage Attack in Federated Learning." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR-2023), pp. 3974-3983. [\[Article\]](#)
- *Atul Sharma*, Pranjal Jain, Ashraf Mahgoub, Zihan Zhou, Kanak Mahadik, and Somali Chaterji. "Lerna: transformer architectures for configuring error correction tools for short-and long-read genome sequencing." BMC bioinformatics 23, no. 1 (2022): 1-26. [\[Article\]](#)
- AK Mukhopadhyay, *Atul Sharma*, Indrajit Chakrabarti, Arindam Basu, and Mrigank Sharad. "Power-efficient spike sorting scheme using analog spiking neural network classifier." ACM Journal on Emerging Technologies in Computing Systems (JETC) 17, no. 2 (2021): 1-29. [\[Article\]](#)

ACADEMIC PROJECTS

Personalization in Decentralized Learning [Under review at ICLR 2026]

Purdue University

Prof. Saurabh Bagchi, Prof. Somali Chaterji, Prof. Chaoyue Liu

Aug 2025 - Present

- Designed a decentralized P2P protocol that learns latent statistical topology of non-IID networks via social triangulation, where nodes share ranked neighbor lists and train per-node embeddings with a ladder triplet loss.
- Achieved personalization matching validation-based baselines without a server, validation data, or cluster priors, using a four-slot explore-exploit sampler and a tent-weighted warmup anchor for stable affinity estimation.

Tiered Gossip Learning [Accepted at NeurIPS 2026]

Purdue University

Prof. Saurabh Bagchi, Prof. Somali Chaterji, Prof. Chaoyue Liu

Jan 2023 - May 2026

- Innovated a hybrid network topology, Tiered Gossip Learning (TGL), a novel two-layer push-gossip-pull protocol that achieves dense global model mixing and scalability by offloading aggregation to a dynamic relay backbone.
- Demonstrated through theoretical analysis and empirical validation that TGL achieves matching convergence rates to decentralized local SGD while reducing communication costs by up to 80% on non-IID vision and language tasks.

Byzantine-robust Federated Learning [\[Article\]](#) [\[GitHub\]](#)

Purdue University

Prof. Saurabh Bagchi, Prof. Somali Chaterji, Prof. Qiang Qiu

Mar 2020 - Aug 2022

- Developed a byzantine-robust aggregation algorithm for federated learning that defends against state-of-the-art model poisoning attacks based on a novel statistical metric called the flip-score to identify malicious client behavior.
- Demonstrated the robustness of the defense over a wide range of scenarios including multiple machine learning models and datasets even in the presence of an adaptive whitebox attacker with full knowledge of the defense.

Automated Tuning of Genomic Error Correction Tools Using Language Models

Purdue University

Prof. Somali Chaterji [\[Article\]](#) [\[GitHub\]](#)

Aug 2019 - Jun 2020

- Architected **Lerna**, a framework that **adapts non-directional Transformer architectures** for automated genomic tool configuration, delivering an **18x speedup** in inference time compared to state-of-the-art RNN models.
- Eliminated the need for a reference genome by leveraging the **perplexity metric** as a fitness measure, maintaining an alignment rate within **0.31% of the theoretical optimum** across diverse Illumina and Nanopore datasets.

RELEVANT COURSES

Data Security and Privacy, Security Analytics, Distributed Computing Systems, Computer Network Systems, Fault Tolerant Computer System Design, Artificial Intelligence, Operating Systems, Machine Learning, Deep Learning